

FINANCIAL ANALYSIS OF
ADVANCED MICRO DEVICES, INC. AND INTEL CORPORATION

Prepared for

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Decoding of Corporate Financial Statements

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INTRODUCTION

The semiconductor industry is at the core of technological advancement, powering devices and applications that define life today. This report focuses on Advanced Micro Devices (AMD) and Intel Corporation, two significant players whose competition has influenced the industry's development. Comparing these businesses' financial results, operational effectiveness, and strategic positioning between 2020 and 2023 will shed light on their achievements and difficulties.

AMD has made great strides in recent years, challenging Intel's long-standing dominance with its Ryzen processors and EPYC server solutions. On the other hand, as market conditions have changed, Intel has come under increasing pressure to hold onto its top position. By examining their financial statements and profit margins, return on assets (ROA), and return on common equity (ROCE), this research investigates how well each business uses its resources, controls costs, and distributes profits to shareholders. AMD and Intel work in a capital-intensive sector where maintaining competitiveness requires large investments in R&D. This report explores how these investments affect their capacity for innovation and financial stability. The report also looks at the connection between reported net income and cash flow from operations to shed light on the nature of their earnings.

The comparison seeks to answer critical financial questions, such as how AMD and Intel's profitability, operating costs, and revenue growth compare over time. Which business maximizes shareholder value through controlling expenses and using assets more effectively? This research assesses organizations' operational effectiveness and financial health by concentrating on critical financial measures, including profit margins, ROA, and ROCE. This analysis makes it easier to understand how AMD and Intel's economic strategies have influenced their competitive positions and the implications of these findings for their futures in the semiconductor business.

EXECUTIVE SUMMARY

Intel Corporation

Intel Corporation, founded in 1968 and headquartered in Santa Clara, California, is a global leader in semiconductor innovation and manufacturing. Intel develops and produces advanced computing technologies, including central processing units (CPUs), memory solutions, networking components, and specialized hardware for artificial intelligence and edge computing. Its flagship products, such as the Intel Core and Xeon processors, are foundational to various applications, including personal computing, enterprise servers, cloud data centers, and Internet of Things (IoT) devices.

Intel is classified as a semiconductor and technology company, recognized for its dominant role in the global CPU market and its leadership in driving advancements in computing technologies. The company's commitment to research and development and its robust manufacturing capabilities position it as a pivotal player in the evolution of high-performance

computing, connectivity, and artificial intelligence. Intel continues to shape the future of technology through its focus on innovation, sustainability, and strategic growth initiatives.

Advanced Micro Devices, Inc.

Advanced Micro Devices (AMD) is a multinational semiconductor company founded in 1969 and headquartered in Santa Clara, California. AMD designs and manufactures high-performance computing and graphics solutions, including central processing units (CPUs), graphics processing units (GPUs), and chipsets. These products are widely utilized in personal computers, gaming consoles, data centers, and embedded systems, serving both consumer and enterprise markets. Known for its innovative designs, such as the Ryzen CPUs and Radeon GPUs, AMD has established itself as a competitive force in the semiconductor industry, offering products with advanced performance, energy efficiency, and value.

AMD is classified as a semiconductor and technology company specializing in hardware solutions for computing and graphics. It has earned recognition as a disruptor in the market, directly challenging industry leaders like Intel and NVIDIA. AMD has positioned itself as a leader in high-performance computing, gaming, and cloud-based solutions, reinforcing its reputation as a cutting-edge technology provider.

AMD is well positioned to compete with companies such as Intel and NVIDIA through its robust, high-tech product offerings and significant cash flow. AMD has found it challenging to breakthrough in the new generation of AI chips but enjoys a healthy share of the personal computer and gaming chip market. We believe that AMD has good financial foundations and will improve despite recent slips.

INCOME STATEMENT ANALYSIS – INTEL (2020-2023)

Please see Appendix A for Income Statement Analysis Figures.

Revenue Growth and Cyclicity: Intel's revenue grew 1.5% from 2020 to 2021, followed by a 20.2% decline in 2022 and a further 14% decrease in 2023. The modest increase in 2021 can be attributed to the surge in demand for personal computing and data center products during the COVID-19 pandemic. However, the sharp declines in subsequent years reflect the cyclical nature of the semiconductor industry, driven by reduced consumer demand, supply chain disruptions, and intensifying competition from AMD and Nvidia.

Net Income Growth and Competitiveness: Net income fell by 59.6% in 2022 and by 79.1% in 2023. This decline reflects the revenue drop, rising operational costs, and pricing pressures. As a 7-8% average growth in NI indicates competitiveness for mature companies, Intel's performance falls significantly short of this benchmark, raising concerns about its ability to maintain profitability in a competitive market.

Key Observation: Intel's inability to sustain net income growth aligns with industry trends where competition, such as AMD's advancements in processor technology, exerts pricing pressure. Additionally, macroeconomic effects, such as reduced consumer spending and inventory build-up, compounded the decline.

Cost of Goods Sold (COGS) and Profit Margin: COGS saw an increase in 2021 and 2022 and a 10.1% decline in 2023. Profit margins consistently fell from 2020 to 2023, reflecting a declining ability to generate profit from sales. The initial rise in COGS and declining profit margins indicate Intel's struggles to manage rising production costs and intense competition.

Key Insight on Value: While Intel's Profit Margin continuously declined in 2021 and 2022, COGS increased, leading to a disproportionate impact on profitability. Fixed production costs remained high despite lower production volumes, showing the challenges in scaling down operations during a downturn and maintaining cost efficiency while innovating at scale. Also, Intel's significant capital expenditures for building new fabs and expanding manufacturing capacity (aligned with Moore's Law) added to fixed costs and depreciation, further weighing on margins.

Growth in Sales vs. Net Income: The decline in sales in 2022 and 2023, respectively, were accompanied by even steeper declines in NI.

Analysis of Cyclicality: This correlation shows the cyclical nature of Intel's business, where external economic conditions and market demand heavily influence revenue and profitability. However, deviations may arise due to internal factors like production inefficiencies or shifts in product mix.

Linked Patterns: The patterns appear correlated when plotting sales and net income changes. As sales fell sharply, NI followed a similar trend. These similar patterns further underscore Intel's cyclical nature, where external market conditions significantly impact both metrics. However, the sharper decline in NI compared to sales highlights the disproportionate impact of declining revenue on profitability due to high fixed costs.

Macroeconomic and Competitive Effects: Intel's performance reflects significant macroeconomic pressures, including supply chain constraints and inflationary costs. Additionally, AMD's competitive advancements in chip technology have pressured Intel's pricing and market share. Companies closer to end consumers (e.g., Intel with personal computing) tend to react more rapidly to macroeconomic changes, intensifying Intel's sensitivity to market conditions.

Accounts Receivable and Sales: When the AR change is higher than sales growth, it indicates an increase in credit terms or delays in collections. The disproportionate growth in AR relative to

sales in 2021 aligns with this observation. This inefficiency impacts Intel's cash flow and raises concerns about financial flexibility. By 2023, the closer alignment of AR and sales trends suggests improvements in collection efforts, though the overall decline in sales underscores persistent macroeconomic and competitive pressures.

INCOME STATEMENT ANALYSIS – AMD (2020-2023)

Please see Appendix A for Income Statement Analysis Figures.

Revenue Growth and Cyclicalities: AMD's revenue grew by 68.3% from 2020 to 2021 and 43.6% from 2021 to 2022, followed by a slight decline in 2023. The significant growth in 2021 and 2022 was driven by strong demand for AMD's Ryzen processors and Radeon graphics cards and the strategic acquisition of Xilinx in 2022, which enhanced its product offerings and market share. The revenue decline in 2023 reflects the cyclical downturn typical in the semiconductor industry, influenced by factors such as market saturation and intensified competition.

Net Income Growth and Competitiveness: Net income rose by 27% in 2021, reflecting operational efficiency during a period of high demand, but fell sharply by 58.3% in 2022 and 35.3% in 2023. The initial increase aligns with revenue growth, while the subsequent declines point to rising costs and pricing pressures as competition intensifies. According to industry benchmarks, an 8% average net income growth indicates competitiveness for mature companies. AMD's inability to sustain this benchmark in 2022 and 2023 raises concerns about its long-term profitability amidst increased competition from Nvidia and Intel.

Key Observation: The steep decline in net income highlights AMD's struggles to balance revenue growth with profitability. Despite its competitive product lineup, higher production costs and pricing pressures significantly impacted AMD's ability to maintain margins.

Cost of Goods Sold (COGS) and Profit Margin: COGS grew in line with sales, but AMD's gross profit margin decreased consistently from 2020 to 2023. This decline reflects challenges in managing production costs and maintaining pricing power despite its position in a high-margin industry.

Key Insight: For the year 2021-2022, COGS increased while Profit Margin decreased. Continued challenges from the global chip shortage and high logistics costs increased production expenses. The complex addition of Xilinx products and costs and increased R&D spending compressed operating income.

Growth in Sales vs. Net Income: AMD achieved exceptional sales growth in 2021 and 2022, far exceeding the “normal” growth threshold of 7%. However, the decline in 2023 was accompanied by a sharper drop in net income, indicating the disproportionate impact of fixed costs and operating expenses on profitability during revenue slowdowns.

Analysis of Cyclicality: The alignment of sales and NI patterns confirms AMD’s cyclical nature, where external demand fluctuations significantly affect financial performance. The sharper decline in net income compared to sales suggests that fixed costs and operational inefficiencies magnify the impact of revenue slowdowns.

Observation on Net Income and Sales correlation: When plotting sales changes alongside net income, the growth in AMD's sales in 2022 and the simultaneous decline in NI reflect the Xilinx acquisition, where it likely introduced one-time charges (e.g., acquisition-related expenses and restructuring costs), which weighed heavily on net income for 2022. Also, the increase in sales in 2022 was driven by the acquisition, which expanded AMD's market share. However, the patterns for NI and Sales for 2023 appear correlated, demonstrating AMD’s cyclical business model.

Macroeconomic and Competitive Effects: AMD's results reflect macroeconomic challenges, including inflationary pressures and reduced consumer spending, impacting product demand. Additionally, Nvidia's dominance in AI and graphics processing units (GPUs) created significant competitive pressure, while Intel’s aggressive investments in chip manufacturing further eroded AMD’s market share in key segments.

Accounts Receivable and Sales: The relationship between AR and sales highlights AMD’s ability to manage credit efficiently. However, in 2023, AR increased, and sales decreased. A rising AR balance with falling sales can strain a company's cash flow as the money customers owe is not readily available.

INCOME STATEMENT – COMPARISONS, OBSERVATIONS, AND RELATIONSHIPS

Revenue Growth: AMD outperformed Intel in revenue growth from 2020 to 2022 due to aggressive market share gains and the acquisition of Xilinx. Nevertheless, both companies experienced revenue declines in 2023 due to the cyclical downturn in the semiconductor industry.

Net Income/Sales Trends: Both companies exhibited a strong alignment between sales and NI trends, underscoring their cyclical nature. AMD’s superior revenue growth did not translate into stronger NI performance, as cost challenges mirrored those of Intel.

Profit Margins and COGS: AMD maintained lower COGS than Intel, leveraging its reliance on third-party manufacturing. However, both companies saw declining profit margins, indicating industry-wide cost pressures and competitive pricing challenges.

Accounts Receivable: AMD displayed better A/R management, ensuring cash inflows aligned more closely with sales. Intel's higher AR relative to sales in earlier years indicated inefficiencies in cash collection.

While AMD demonstrated stronger revenue growth during upcycles, its declining profitability during downturns mirrored Intel's challenges. AMD's superior A/R management and streamlined cost structure provide an edge in operational efficiency, but its inability to sustain profitability limits its competitiveness. In contrast, Intel's higher fixed costs and operational inefficiencies exacerbate its vulnerability to cyclical pressures. Overall, AMD's income statement reflects a stronger growth and operational efficiency position, while Intel's performance raises concerns about its adaptability and cost management. Investors may view AMD as better positioned for growth, though both companies face challenges navigating cyclical downturns.

BALANCE SHEET ANALYSIS – INTEL (2020-2023)

Please see Appendix B for Balance Sheet Analysis Figures.

Assets

Total Assets: Intel's total assets grew 10% in 2021, 8.13% in 2022, and 5.2% in 2023, reflecting consistent yet slowing growth. These trends align with Intel's substantial investments in PP&E to expand manufacturing capacity, driven by Moore's Law, which demands consistent innovation to avoid obsolescence, driving Intel's aggressive capital expenditures. The slowing growth rate highlights Intel's cautious debt management approach amidst rising interest rates and global economic uncertainties.

Inventory and Inventory Turnover: Inventory turnover declined -9.79% in 2021 and saw an even sharper decline of -17.76% in 2023. This trend shows the inefficiencies in managing stock levels, exacerbated by the need to maintain inventory amid fluctuating demand. As the semiconductor industry moved into a downturn in 2022–2023, inventory built up during the boom period remained unsold, reducing turnover.

Accounts Receivable Turnover: Accounts receivable turnover fell in 2020-2022, signaling slower revenue collection. However, it increased in 2023, indicating that the company is conservative about extending credit to customers and is efficient or aggressive with its collection practices.

Total Asset Turnover (TAT): TAT steadily decreased, underscoring inefficiencies in utilizing assets to generate revenue. These declines are tied to Intel's significant PP&E investments,

which are yet to yield proportional financial returns. This trend highlights the long lead time between investment and operational gains in a capital-intensive industry, such as semiconductors, where production requires advanced machinery and large-scale facilities.

Return on Assets (ROA): ROA saw an important decrease in 2022 of -62.68%, reflecting Intel's declining ability to convert its asset base into profit. This trend highlights the strain of balancing high fixed costs with cyclical revenue patterns, where downturns in demand disproportionately impact profitability.

Liabilities

Total Liabilities: Intel's total liabilities grew steadily, increasing by 1.34% in 2021, 10.5% in 2022, and 1.15% in 2023. The slower growth rate reflects Intel's cautious approach to debt management amidst rising interest rates and global economic uncertainty. Intel's reliance on debt financing to support strategic initiatives like new fabs and R&D highlights its dependence on external funding in a capital-intensive industry.

Debt-to-Equity Ratio: The debt-to-capital ratio rose consistently, reflecting Intel's increasing reliance on debt to finance growth. While necessary for funding large-scale projects, this trend increases financial risk, particularly during cyclical downturns when revenue generation slows.

Current Liabilities: Current liabilities steadily rose in 2020, 2021, and 2022. Key contributors include accounts payable, which reflects heightened procurement activity during manufacturing expansions. These rising current liabilities portray the cyclical demand for securing materials during peak production periods and align with Intel's cash management strategy. However, current liabilities decreased in 2023. This indicates Intel's increasing reliance on short-term obligations to fund operations.

Shareholder Equity

Total Equity: Intel's shareholders' equity grew consistently, increasing by 17.71.% in 2021, 8.28% in 2022, and 6.47% in 2023. Despite the growth, the slowing rate reflects increasing liabilities, declining net income, and continued dividend payouts, which limit reinvestment potential.

Return on Equity (ROE): ROE decreased consistently, highlighting a sharp decline in Intel's ability to generate shareholder returns. The reduction reflects rising operational costs, declining net income, and competitive pressures, especially from AMD and Nvidia.

Return on Common Equity (ROCE): ROCE declined consistently during the 2020-2023 period. This decrease highlights Intel's reduced ability to generate returns on equity attributable to common shareholders. The decline emphasizes the impact of declining profitability, increased debt, and reduced operational efficiency.

Altman Z-Score:

In 2022, Intel moved from the "gray area" into the "high probability of bankruptcy" zone, reflecting significant financial stress. Intel fell behind competitors like AMD and TSMC in advanced fabrication nodes, increasing production inefficiencies. Also, strategic spending to build new fabs and compete with TSMC strained Intel's finances, amplifying the falling revenue and EBIT effects.

BALANCE SHEET ANALYSIS – AMD (2020-2023)

Please see Appendix B for Balance Sheet Analysis Figures.

Assets

Total Assets: AMD's total assets experienced rapid growth, surging by 38.57% in 2021, 444.17% in 2022, and 0.45% in 2023. The extraordinary growth in 2022 reflects AMD's acquisition of Xilinx, contributing to significant goodwill and intangible asset increases. Unlike Intel, AMD's expansion strategy relied more on acquisitions than capital-intensive investments, which is an alternative path to growth within the semiconductor industry. The slower growth in 2023 reflects a stabilization period post-acquisition.

Inventory and Inventory Turnover: Inventory rose from 2020 to 2021 but declined from 2021 to 2023, reflecting inefficiency in managing stock levels as inventory surged in response to fluctuating demand and the Xilinx integration. The slower turnover shows challenges in optimizing inventory levels in a competitive and cyclical industry where technology advancements can render products obsolete quickly.

Accounts Receivable Turnover: Accounts receivable turnover increased from 2020 to 2022. However, it decreased in 2023. The drop aligns with AMD's increased revenue and broader customer base post-acquisition but also suggests potential inefficiencies in cash collection practices, exacerbated by macroeconomic pressures.

Total Asset Turnover (TAT): TAT decreased significantly in 2022, reflecting reduced efficiency in utilizing assets to generate revenue. The decline highlights the impact of the Xilinx acquisition, where adding goodwill and intangibles inflated the asset base, diluting the turnover

ratio. This trend underscores the challenge of integrating acquisitions while maintaining operational efficiency.

Return on Assets (ROA): ROA dropped consistently from 2020 to 2023, emphasizing AMD's declining ability to generate profits from its asset base. However, there was a steep decline of -92.33% , which reflects the combination of lower net income and significant asset growth, driven by the Xilinx acquisition and increased inventory levels.

Liabilities

Total Liabilities: AMD's total liabilities grew by 57.5% in 2021, 160.07% in 2022, and 6.52% in 2023. The sharp increase in 2022 reflects the assumption of Xilinx's debt alongside increased accounts payable and accrued liabilities. The slower growth in 2023 indicates AMD's efforts to stabilize its financial position post-acquisition.

Debt-to-Equity Ratio: AMD's debt-to-capital ratio increased significantly in 2022 due to the acquisition. However, it decreases in 2023, smoothing the effects on debt post-acquisition. While the ratio remains lower than Intel's, the sharp rise increases AMD's vulnerability to cyclical downturns and interest rate hikes.

Current Liabilities: Current liabilities grew by 75.42% in 2021, 50.21% in 2022, and 5.02% in 2023. The rise reflects increased accounts payable and accrued liabilities linked to production ramp-ups and integration costs for Xilinx. Despite the increase, AMD maintained a strong current ratio, indicating sufficient liquidity for short-term obligations.

Shareholder Equity

Total Equity: AMD's shareholders' equity experienced robust growth, increasing by 28.67% in 2021, 630.29% in 2022, and 2.09% in 2023. The 2022 spike reflects the all-stock acquisition of Xilinx, which significantly boosted equity through additional paid-in capital. The slower growth in 2023 reflects AMD's stabilization efforts and the impact of declining net income.

Return on Equity (ROE): ROE decreased from 2020-2023, with a significant decrease of -94.28% in 2022. This reflects AMD's challenges in generating shareholder returns amidst rising costs and competitive pressures. The decline emphasizes the impact of slowing revenue growth and the dilution of equity post-acquisition.

Return on Common Equity (ROCE): While ROCE increased by 113.24% in 2021, it dropped -95.37% in 2022. This most likely signals that the acquisition is not immediately generating sufficient returns relative to the capital invested.

Altman Z-Score

AMD consistently moved from the “gray area” into the “low probability of bankruptcy” zone in 2021. Also, even though it saw a dip in 2022 due to the acquisition, AMD increased its Z score in the subsequent year. Despite the industry’s cyclical nature, AMD maintained robust revenue growth and cost control, reflecting strong operational health supported by a strong equity base and strategic investments.

BALANCE SHEET – COMPARISONS, OBSERVATIONS, AND RELATIONSHIPS

Asset Efficiency: Intel's declining ROA and TAT reflect inefficiencies in leveraging assets for revenue, which are linked to heavy PP&E investments. AMD's similar declines in 2022 and 2023 are driven by the dilution effect of goodwill and intangibles post-Xilinx acquisition.

Cyclical Trends: Both companies faced declining inventory turnover in 2022, highlighting exposure to cyclical demand patterns. Intel’s improvement in 2023 contrasts with AMD’s continuing challenges post-acquisition.

Debt Management: Intel’s higher debt-to-capital ratio reflects greater reliance on external financing for manufacturing expansions. AMD’s lower ratio, despite rising post-acquisition, demonstrates a more conservative leverage approach.

Shareholder Value: Declining ROE, ROCE, and Altman Z-scores for both companies highlight financial strain, particularly in 2022. However, AMD’s stronger Z-score reflects higher financial stability, while Intel’s proximity to the “gray area” signals greater vulnerability.

Both Intel and AMD demonstrate strengths and weaknesses in their balance sheets. Intel’s capital-intensive strategy positions it for long-term growth but reveals inefficiencies in asset utilization and declining profitability. AMD’s acquisition-driven growth has boosted its asset base and equity but poses integration and efficiency challenges. AMD appears more favorable for immediate financial stability with a higher Altman Z-score and lower reliance on debt. However, Intel’s focused investments could yield long-term advantages if managed effectively. Investors must weigh risk tolerance and time horizon in evaluating these companies.

STATEMENT OF CASH FLOW ANALYSIS – INTEL (2020-2023)

Please see Appendix C for Statement of Cash Flows Analysis Figures.

Operating Activities

Intel’s cash flow from operations (CFO) consistently declined over the period. These trends indicate growing inefficiencies in converting operational performance into cash, a critical metric for a cyclical, capital-intensive company. The accruals (NI from Ops - CFO) rose steadily,

signaling increased reliance on uncollected revenues. High accruals pose risks, reflecting a potential inability to convert reported earnings into tangible liquidity. This is particularly concerning during periods of heightened competition and cyclical downturns.

The decline in CFO mirrors challenges in managing assets, particularly working capital. Inventory levels surged between 2020 and 2022, driven by stockpiling ahead of product launches and global supply chain disruptions. These inefficiencies underscore Intel's struggle to adapt to intensifying competition from AMD and shifting market conditions. The declining CFO also indicates an overreliance on accrual-based income during a period when operational cash generation is vital to fund investments.

Investing Activities

Intel's CFI showed significant outflows, reflecting aggressive capital expenditure trends. CFI decreased in 2021 as Intel ramped up investments in PPE under the IDM 2.0 strategy. In 2022, CFI increased by 57.9%, reflecting Intel's continued focus on high-capital projects and investments. This shows Intel's prioritization of future growth over immediate liquidity. By 2023, CFI decreased sharply by -135.0%, as Intel reduced its capital expenditures in response to ongoing revenue declines and cash flow pressures, shifting towards more cautious financial management.

The mismatch between Intel's PPE investments and depreciation continues to raise concerns. Capital investments should align with depreciation to maintain operational competitiveness without overextending resources. Intel's substantial outlays have strained its liquidity, leading to reliance on asset sales, such as marketable securities, as supplementary funding sources.

The significant negative free cash flow (CFO—Fixed Assets Acquired) in 2023 shows Intel's struggle to balance aggressive capital spending with financial sustainability. While such investments align with the cyclical nature of the semiconductor industry, the reliance on external funding and strained cash reserves highlight Intel's vulnerability during downturns. This negative trend raises questions about the sustainability of Intel's aggressive capital spending.

Financing Activities

In 2022, CFF increased sharply by 119%, turning positive as Intel relied on substantial short-term borrowing to fund its IDM 2.0 expansion initiatives and maintain flexibility amid the semiconductor industry's cyclical challenges. This reliance on external funding reflects Intel's strategy to support aggressive capital expenditures while navigating competitive pressures from AMD and Nvidia. By 2023, CFF surged further, increasing by 662.9%. This sharp rise shows Intel's pivot toward leveraging external financing to stabilize its liquidity position and ensure long-term strategic resilience.

While the consistent increases in CFF reflect Intel's ability to secure external funding, it also portrays growing financial risk. Though strategic in a capital-intensive industry, reliance on debt raises concerns about Intel's capacity to manage debt obligations effectively during cyclical downturns. The focus on maintaining dividend payouts during the rising financial strain demonstrates Intel's commitment to shareholder returns. Still, questions remain about the sustainability of such strategies in the face of declining operational cash flows.

STATEMENT OF CASH FLOW ANALYSIS – AMD (2020-2023)

Please see Appendix C for Statement of Cash Flows Analysis Figures.

Operating Activities

AMD's cash flow from operations (CFO) showed consistent growth initially but experienced volatility as the company navigated a cyclical industry and strategic shifts. CFO increased by +229% in 2021, reflecting operational improvements amid strong semiconductor demand. This robust growth was followed by a slight increase, but it further dropped by -53.2% in 2023. The sharp declines in later years highlight challenges in sustaining operational efficiency during periods of slowing demand.

The rising accrual ratio (Net Income from Operations - CFO) throughout the period signals increased reliance on uncollected income, underscoring inefficiencies in working capital management. Contributing factors include elevated inventory levels and slower accounts receivable collections, which pressured cash flow. By 2023, AMD's declining CFO reflected broader market trends of weakening demand and intensifying competition within the semiconductor industry. Despite these challenges, AMD's CFO remained positive throughout the period, demonstrating the company's resilience in generating operational cash, which supported liquidity during an uncertain market environment.

Investing Activities

AMD's CFI activities were dominated by the 2022 acquisition of Xilinx, which significantly reshaped its balance sheet. CFI outflows increased by +27.9% in 2021 as AMD ramped up investments in PPE to enhance production capabilities. However, in 2022, CFI surged by +391.4%, reflecting the \$35 billion all-stock Xilinx acquisition. This acquisition brought substantial goodwill and intangibles to AMD's balance sheet, reshaping its competitive position by expanding its capabilities in programmable logic devices.

By 2023, CFI normalized, with outflows decreasing by -171.2% compared to 2022. This reduction in investment outflows signaled a stabilization phase after the aggressive expansion of the prior year. However, the mismatch between depreciation and investments raises concerns about AMD's ability to maintain its operational competitiveness over the long term. The sharp increase in CFI during 2022, largely funded by financing activities, underscores the strain on AMD's liquidity from such a large acquisition.

Financing Activities

By 2021, CFF shifted significantly to a negative outflow, decreasing by -31,583.3%, as AMD repurchased shares to support its stock price and returned capital to shareholders. In 2022, CFF outflows intensified, dropping by 72.2%, as AMD partially funded the \$35 billion Xilinx acquisition through equity issuance and debt financing. This reflects a period of heavy financial strain as the acquisition realigned AMD's strategic focus toward expanding its capabilities. The sharp drop underscores AMD's reliance on external financing to sustain its competitive edge in the semiconductor market. However, in 2023, CFF outflows significantly improved, increasing by +64.9%, as AMD pivoted toward stabilizing its capital structure by reducing reliance on external debt and improving financial flexibility. This contrasts with Intel's more aggressive

reliance on short-term borrowing and dividend payouts during the same period, emphasizing AMD's prioritization of reinvestment and long-term growth over immediate shareholder returns.

Data Source: Stock Analysis & Edgar Library (retrieved Dec 3, 2024)

STATEMENT OF CASH FLOWS – COMPARISONS, OBSERVATIONS, AND RELATIONSHIPS

Assets and CFO: Intel and AMD both experienced rising accruals, reflecting inefficiencies in converting operational income into cash. While AMD's CFO remained positive throughout 2020-2022, Intel's CFO declined significantly, reflecting greater challenges in working capital management. Both companies faced pressures from elevated inventory levels, but AMD's accruals were less pronounced, showcasing better operational cash flow management.

Liabilities and CFF: Both Intel and AMD increased their reliance on debt in 2022 to fund aggressive capital expenditures. Intel used short-term borrowing to fund its IDM 2.0 strategy, while AMD issued debt to support the Xilinx acquisition. In 2023, both companies shifted focus to deleveraging, though AMD's approach was more balanced, as it avoided dividend payouts and maintained stronger cash reserves.

Cyclical Investments: Both companies exhibited sharp increases in CFI during 2022, driven by aggressive capital investments. Intel focused on expanding PPE, while AMD concentrated on acquiring Xilinx. These trends align with the industry norm of capital spending surges during downturns, though AMD's reduction in outflows by 2023 indicates better stabilization compared to Intel.

Shareholder Impacts: Intel maintained dividend payouts throughout the period, signaling shareholder confidence but straining liquidity. AMD avoided dividend payouts, preserving cash for reinvestment and supporting long-term growth. AMD's approach reflects a more cautious strategy, aligning with its focus on maintaining financial flexibility.

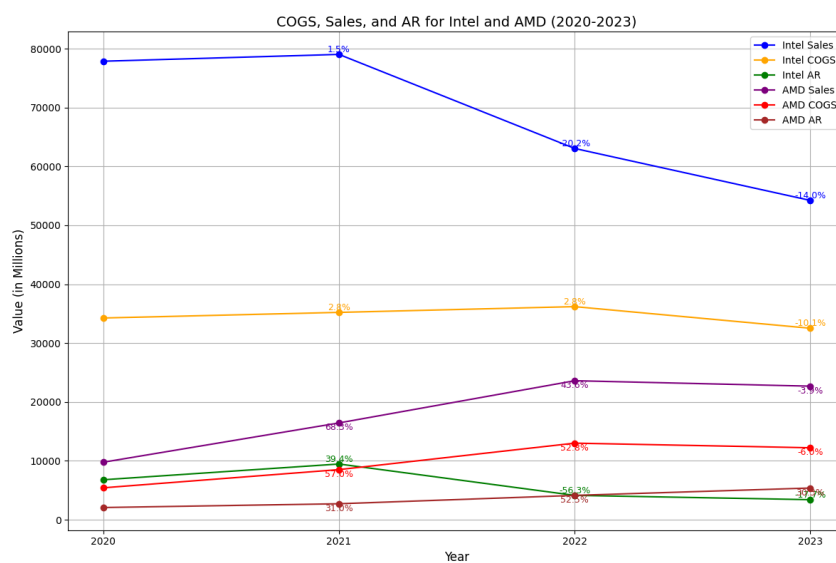
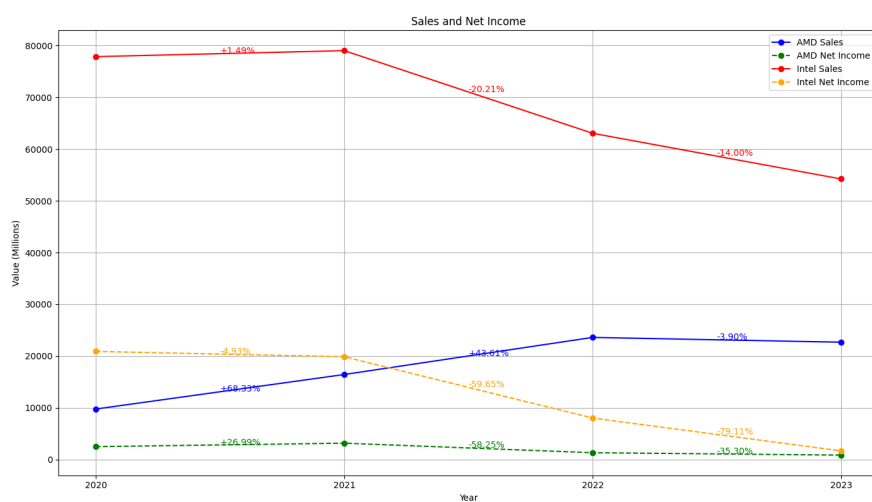
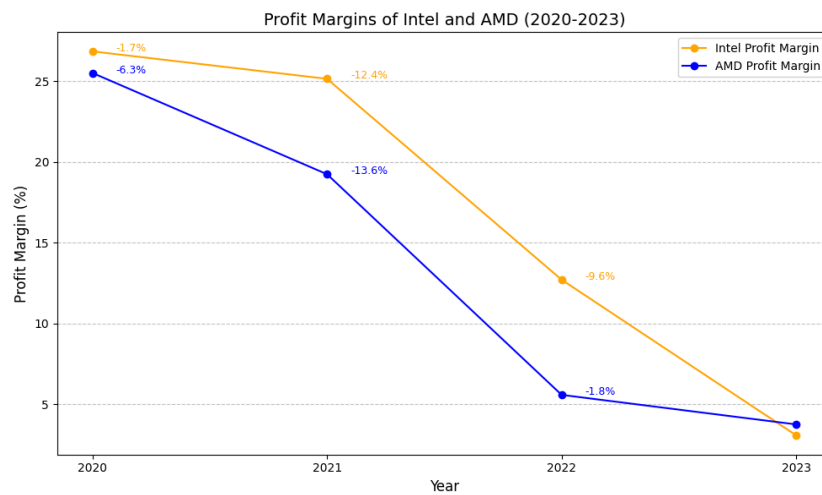
AMD emerges as the stronger performer in cash flow management during the 2020–2023 period. Despite volatility, AMD maintained a positive CFO and demonstrated greater alignment between operational cash flows and investments. The Xilinx acquisition significantly reshaped its financial profile, but AMD's cautious liquidity management and reinvestment strategy positioned it well for long-term growth.

In contrast, Intel's declining CFO and negative free cash flow highlight inefficiencies in cash flow generation and reliance on external funding. While Intel's aggressive investments position it for potential future gains, its short-term financial health appears weaker than AMD's. Based on cash flow performance alone, AMD would likely be considered the more attractive investment option during this period.

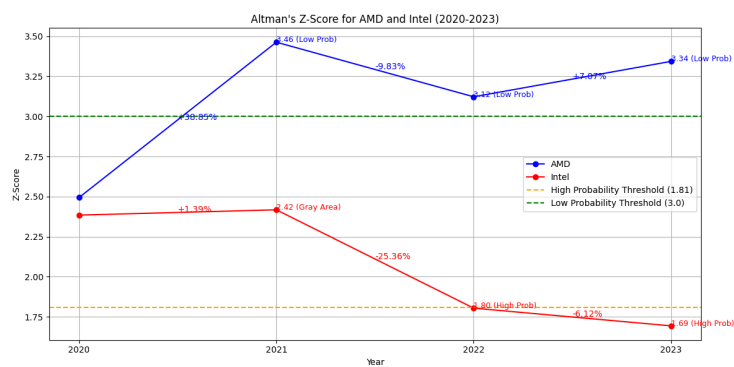
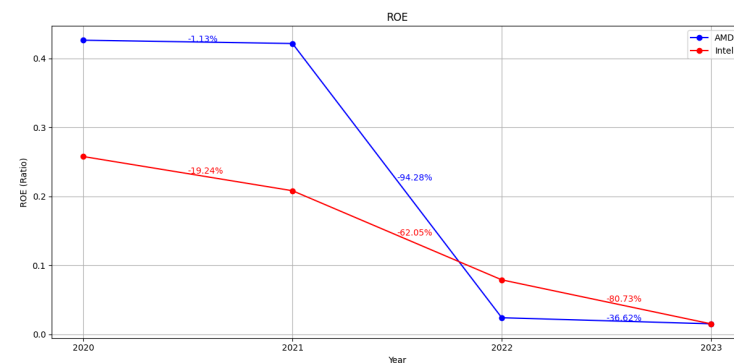
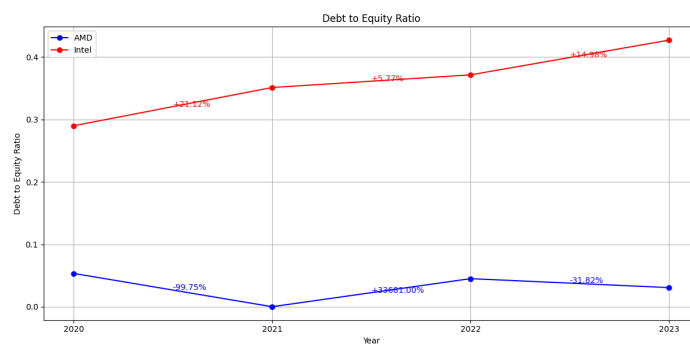
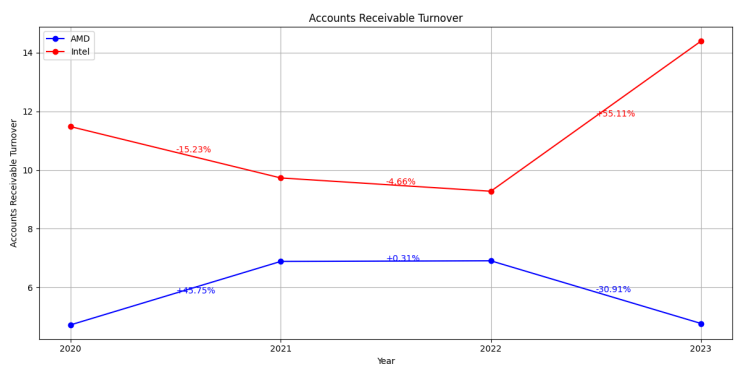
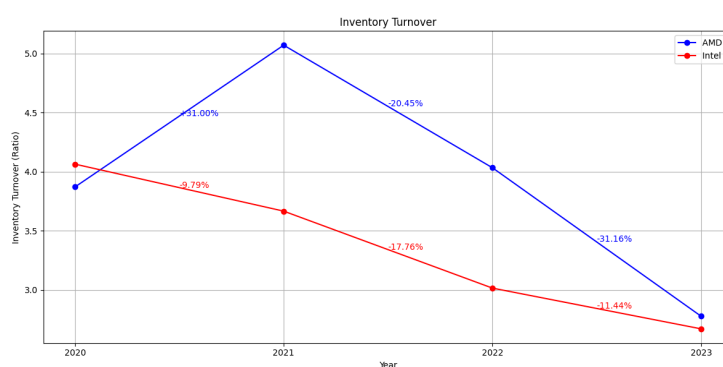
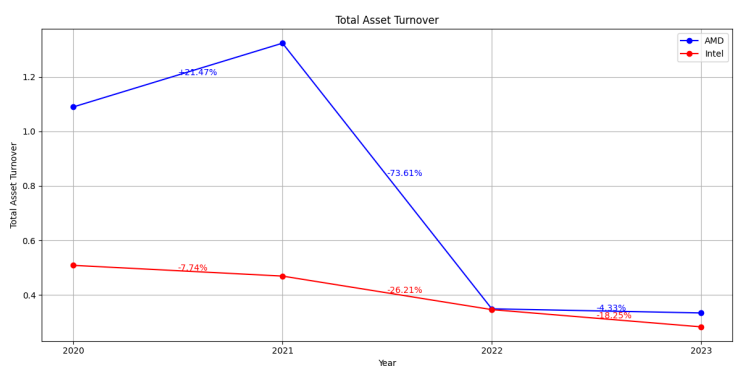
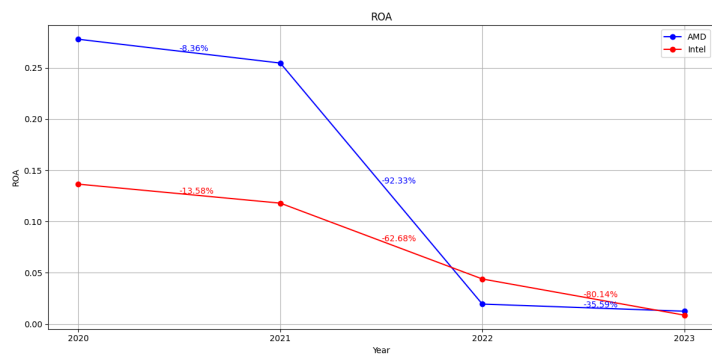
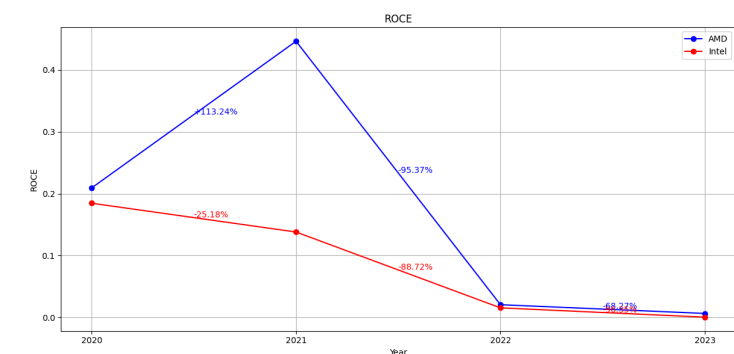
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Appendix A: Income Statement Figures



Appendix B: Balance Sheet Figures



Appendix C: Statement of Cash Flows Figures

